

The effect of cognitive task complexity on gait stability in adolescents following concussion

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Abstract Concussion has been reported to result in disturbances to motor and cognitive functions. One way to examine these disturbances is through a dual-task assessment. Many secondary cognitive tasks have been proposed as appropriate tools during concussion assessment; however, task complexity has not been compared within a dual-task investigation. The purpose of this study was to prospectively examine how gait balance control was affected by three secondary cognitive tasks of varying complexity following concussion. Forty-six adolescents completed a dual-task walking protocol which included walking without any cognitive task (WALK), walking while completing a single auditory Stroop (SAS), multiple auditory Stroop (MAS), and a question and answer task (Q&A). Those who sustained a concussion ($n = 23$, mean age 15.4 ± 1.3 years) reported to the laboratory within 72 h of injury and in the following time increments: 1 week, 2 weeks, 1 month, and 2 months post-injury. Twenty-three healthy control subjects (mean age 15.4 ± 1.3 years), individually matched to each concussion subject, completed the same protocol in similar time increments. The concussion group demonstrated greater total center of mass (COM) medial/lateral displacement in the MAS and Q&A conditions compared with the control group. The concussion group also displayed the greatest peak COM anterior velocity in the least complex condition (WALK), and a significant decrease was observed as task complexity increased (SAS > MAS > Q&A). These

findings indicate that gait balance control may be affected by task complexity following concussion and represent a way to identify motor recovery following concussion.

Keywords Brain injury · Dual task · Gait · Balance · Recovery

Introduction

The management of concussion, particularly related to concussion in sport, has recently become a dynamically evolving area of interest to researchers and clinicians alike. As a result of the complexity of this injury, different medical and sport agencies have recently released evidence-based position statements regarding the scope, pathophysiology, diagnosis, and follow-up care of concussion (Harmon et al. 2013; Giza et al. 2013; McCrory et al. 2013). Although isolated cognitive evaluations have been traditionally thought to be a key element in concussion management (Guskiewicz et al. 2004), it has been suggested that such assessments should never be the sole basis of concussion management, but rather as aids in the clinical evaluation (McCrory et al. 2013). Balance and gait testing have been increasingly identified as available tools documented to aid in the identification of how individuals recover from concussion (Harmon et al. 2013) and as integral components of a comprehensive evaluation (McCrory et al. 2013). Thus, an evaluation which simultaneously assesses cognitive and motor functions may constitute an effective assessment technique for the monitoring of recovery from concussion.

Many investigations have simultaneously assessed motor and cognitive functions following brain injury through the use of a dual-task paradigm (Parker et al. 2005; McCulloch et al. 2010; Fait et al. 2013; Howell et al. 2013b). These

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types of evaluations can measure motor function while performing a cognitive task or vice versa. Using dual-task experiments, investigators have identified deficits in individuals who sustained a concussion for a time of approximately 1 month (Parker et al. 2006; Fait et al. 2013) to many years (Martini et al. 2011) following injury.

While concussion has been observed to be a multi-dimensional injury which affects body movement control and cognitive processes when executed simultaneously (Sosnoff et al. 2008), no investigations to this point have examined how secondary tasks of varying complexity affect the motor performance of individuals acutely suffering from concussion. Adolescents, in particular, may be particularly vulnerable to the effects of concussion due to the development of frontal structures of the brain during this time of life (Luna et al. 2004). Previous experiments have identified deficits still present at 2 months after concussion in a cohort of adolescents in their conflict resolution, task switching, and dual-task gait balance control abilities (Howell et al. 2013a, b).

Dual-task assessments necessitate proper distribution of attentional resources, thus challenging the brain to effectively execute both tasks simultaneously (Yogev-Seligmann et al. 2008). This type of testing represents a way to observe performance differences of a motor task with and without the simultaneous performance of cognitive tasks. Two common cognitive tests that have been reported as effective secondary tasks when introduced simultaneously during motor performance have been a question and answer task, such as continuous subtraction (Catena et al. 2007) and the Stroop task, which has recently been reported to be an appropriate tool in the follow-up care of concussion (Teel et al. 2013) and commonly used when monitoring the effects of sport-related concussion (Vallée et al. 2006; Teel et al. 2013; Fait et al. 2013). Experiments using functional magnetic resonance imaging (fMRI) indicated that the Stroop test activates a focal region of frontal lobe structures, such as the dorsal anterior cingulate gyrus (Bush et al. 1998) or the prefrontal cortex (Fan et al. 2003).

Question and answer tasks, such as reciting the months in reverse order, have been utilized in the sideline care of sport concussion via standardized concussion assessment tools (McCrory et al. 2009). fMRI investigations have revealed that arithmetic tasks engage a high number of cognitive processes as well as activation in the parietal areas of the brain and other brain areas such as the fusiform and lingual gyri, Brodman's area, the prefrontal cortex, and thalamus (Rickard et al. 2000). Spelling tasks have similarly been observed to activate a complex network of brain structures, including the left inferior frontal gyrus, left middle/superior frontal gyrus, left superior parietal lobe, and left fusiform gyrus (Purcell et al. 2011). These investigations suggest that question and answer

tasks require multiple brain area activation for successful completion.

Accordingly, a Stroop task which is performed only once during a testing session represents a less complex task than one performed multiple times in succession during a similar session. An increase in complexity between two such conditions is a result of the necessary increase in sustained attention for completion. A question and answer task represents a more complex condition than a multiple Stroop condition as it possesses similar temporal demands with an increased integrative requirement of multiple brain structures to coordinate and execute such a task.

Dual-task paradigms are commonly used in the recovery monitoring and management of concussion suggesting that protocols involving simultaneous attention to cognitive and dynamic/static postural stability may reveal subtle post-concussion disturbances better than single-task paradigms (Resch et al. 2011; Ross et al. 2011). However, to our knowledge, no current literature exists which directly compares the effects of secondary tasks of differing complexity on gait performance. Therefore, the purpose of this study was to prospectively examine how gait balance control is affected by secondary cognitive tasks of varying complexity implemented simultaneously while walking in a population of adolescents with and without concussion. It was hypothesized that (1) secondary tasks of greater complexity would result in greater disturbances to walking balance control than less complex secondary tasks and single-task walking in subjects with concussion and (2) that concussion subjects would be affected by each secondary task to a greater degree than control subjects.

Methods

Participants

Twenty-three high school students (20 males and 3 females) were identified as sustaining a concussion while participating in school sports by a health professional (certified athletic trainer or physician). The definition of concussion was consistent with that of the 3rd International Consensus Statement on Concussion in Sport as an injury caused by a direct blow to the head, face, neck, or elsewhere in the body with an impulsive force transmitted to the head, resulting in impaired neurologic function and acute clinical symptoms (McCrory et al. 2009). Each subject reported to the laboratory for testing within 72 h of injury and returned approximately 1 week, 2 weeks, 1 month, and 2 months post-injury. Each subject who sustained a concussion was then matched with a healthy control subject by sex, height, mass, age, and sport and tested in a similar timeline. Prior to data collection, the institutional review board reviewed and approved

the protocol of the current study. All subjects and parent/guardian (if under the age of 18) provided informed consent. Permission was also granted by each respective school district to conduct testing with student participants.

Each subject who suffered a concussion was removed from physical activity on the day of injury and did not return until cleared by a physician in accordance with state law. Exclusion criteria for all subjects included: (1) lower extremity deficiency or injury which may affect normal gait patterns, (2) history of cognitive deficiencies, such as permanent memory loss or concentration abnormalities, (3) history of three or more previous concussions, (4) loss of consciousness from the concussion lasting more than 1 min, (5) history of attention deficit hyperactivity disorder, or (6) a previously documented concussion within the past year. Consistent with previous work (Howell et al. 2013a), individuals with three or more previous concussions were ineligible to participate in the study to ensure, to the extent possible, the exclusion of subjects experiencing chronic concussion and those who experienced a loss of consciousness for greater than 1 min were excluded because of the role this sign plays in concussion management modification (McCrory et al. 2009). A verbal medical history was taken by a certified athletic trainer for all subjects upon their first visit to the laboratory to confirm all criteria were met for inclusion in the study. Additionally, a symptom inventory was administered to all subjects using a 22-symptom inventory adapted from the SCAT2 (McCrory et al. 2009), which requires subjects to identify the severity of each documented symptom of concussion using a 6-point Likert scale. This inventory indicates the sum of all symptom severity ratings (range 0–132).

Protocol

The protocol for every subject included one single-task and three dual-task walking conditions. For the single-task condition (WALK), the subject walked with undivided attention while barefoot at a self-selected speed along a walkway. For each of the three dual-task conditions, the subject walked along the same walkway but while simultaneously responding to one of three presented cognitive tasks. The order of presentation for the three cognitive tests changed for each of the five testing sessions (72 h, 1 week, 2 weeks, 1 month, and 2 months) but was consistent for all subjects. The three cognitive tasks, in order of complexity, were: (1) a single auditory Stroop (SAS), (2) a multiple auditory Stroop (MAS), and (3) a question and answer task (Q&A). The SAS task required the subject to listen to the words “high” or “low” played one time per trial by computer speakers in a high or low pitch and then identify the pitch of the word, regardless of whether the pitch was congruent with the meaning. The MAS tasks were similar to the

SAS, except the words “high” or “low” were played and the subjects were required to respond four times per trial. Each “high” or “low” stimulus after the first was triggered by the previous response of the subject and was presented one second following the subject response. The Q&A consisted of spelling a five-letter word backwards, subtracting by 6s or 7s, or reciting the months in reverse order. Subjects completed one of these tasks during each trial. The type of Q&A task completed was randomly selected for each trial to avoid learning effects from one trial to the next. The subjects completed 8–10 trials for each of the single- and dual-task conditions.

During dual-task walking, subjects were not instructed to focus on either the cognitive or walking task, but to continue walking while correctly completing the task. For the SAS and MAS conditions, the stimuli were presented once subjects had reached steady-state gait, several steps after gait initiation, and were triggered by a photocell. The instructions for the Q&A task were given to the subjects prior to gait initiation after which they were instructed to begin walking and respond to the task once an auditory beep was played, triggered by a photocell located several steps after gait initiation. For each walking trial, data were analyzed for one gait cycle, defined as heel strike to heel strike of the same limb. The targeted stride for analysis was the one immediately after initial task-stimulus presentation.

A total of 29 retro-reflective markers were placed on bony landmarks of the subject, and whole body movement was recorded using a ten camera motion analysis system (Motion Analysis Corp., Santa Rosa, CA, USA) at a sampling rate of 60 Hz to capture and reconstruct the three-dimensional trajectory of each marker. Marker trajectory data were low-pass filtered using a fourth-order Butterworth filter with a cutoff frequency set at 8 Hz. External markers and estimated joint centers were used to calculate the center of mass (COM) of each individual body segment. Whole body COM position data were then calculated as the weighted sum of all body segments (Winter 2009) with thirteen segments representing the whole body (head/neck, trunk, pelvis, two arms, two forearms/hands, two thighs, two legs, and two feet). Gait events were detected from ground reaction forces collected at 960 Hz using three forceplates (Advanced Mechanical Technologies Inc., Watertown, MA, USA). Both SAS and MAS stimulus was presented in random order using Super Lab Pro (Cedrus Corp., San Pedro, CA, USA). Verbal responses to each task were recorded using a headset wireless system with a microphone (AKG Acoustics, Northridge, CA, USA).

Data analysis

Linear COM velocity was calculated using the cross-validated spline algorithm from COM position (Woltring

1986). The total medial/lateral COM displacement, peak anterior COM velocity, and peak medial/lateral (M/L) COM velocity were calculated and identified during the gait cycle. Peak M/L and anterior COM velocities occurred during the heel strike event of the gait cycle. These variables have previously been used to provide sensitive detection of gait imbalance in elderly individuals (Hahn and Chou 2003) and in those suffering from concussion (Parker et al. 2006). Average walking speed was also calculated as the mean forward velocity during the gait cycle. The mean of the 8–10 trials for each subject was calculated for each variable.

Cognitive task accuracy was recorded by the experimenter during the testing session while subjects completed walking trials. Accuracy was analyzed for each of the three conditions as the total number of correct responses divided by the total number of responses completed during all walking trials.

Statistical analysis

Group demographic differences and testing time differences were tested using an independent *t* test for height, mass, age, and testing day. Three-way mixed effects analyses of variance were used to analyze each dependent variable to determine the interactions and main effects of group (concussion and control), time (72 h, 1 week, 2 weeks, 1 month, and 2 months), and task (WALK, SAS, MAS, Q&A). For all omnibus tests, significance was set at $p < .05$. Follow-up pairwise comparisons were then examined using the Bonferroni procedure to control Family Wise Type I Error. All statistical analyses were performed with Statistical Package for the Social Sciences version 20 (SPSS Inc., Chicago, IL, USA).

Results

Subjects

Demographic and injury data for the subjects groups are presented in Table 1. No significant differences were detected between groups for age ($p = .494$), height ($p = .793$), or mass ($p = .549$). The concussion group included 15 football athletes, 4 soccer athletes, 3 wrestling athletes, and 1 volleyball athlete. The control group included 15 football athletes, 3 soccer athletes, 3 wrestling athletes, 1 volleyball athlete, and 1 basketball athlete.

The concussion group was initially tested at a mean of 2 ± 0.7 days after sustaining a concussion and returned for testing at means of 8 ± 1.7 , 17 ± 3.4 , 31 ± 4.1 , and 59 ± 3.4 days after injury. The control group was similarly tested at means of 8 ± 2.0 , 16 ± 4.5 , 31 ± 3.6 , and

Table 1 Demographic and injury history information for both groups of subjects

	Concussion	Control
Demographics		
Age (years)	15.4 ± 1.3	15.7 ± 1.3
Sex (M/F)	20/3	20/3
Height (cm)	173.7 ± 6.1	173.2 ± 7.8
Mass (kg)	73.0 ± 16.1	70.4 ± 12.7
Injury characteristics		
Loss of consciousness lasting 30 s or less (<i>n</i>)	4	N/A
History of concussion >1 year prior to initial testing day (<i>n</i>)	5	1

57 ± 6.0 days after the initial testing session. No significant differences in testing times for each assessment were detected between the groups ($p > .30$). Similar to previous work (Howell et al. 2013a), evaluation of clinical symptom severity scores indicated a significant group $\times$ time interaction [$F(4, 176) = 15.64$, $p < .001$, $\eta_p^2 = .262$]. Follow-up pairwise tests indicated a significant difference between the concussion and control groups (mean $\pm$ SD) at the 72-h time point ($p < .001$, concussion = 41.6 ± 24.1 , control = 4.5 ± 4.8), 1-week time point ($p < .001$, concussion = 28.4 ± 23.5 , control = 3.3 ± 5.1), and 2-week time point ($p = .004$, concussion = 19.7 ± 20.9 , control = 4.4 ± 9.2). Differences were no longer detected between groups at the 1-month ($p = .060$, concussion = 11.8 ± 13.6 , control = 4.2 ± 6.9) or 2-month testing time points ($p = .308$, concussion = 8.7 ± 17.4 , control = 4.4 ± 6.3).

Total M/L COM displacement

Analysis of the total M/L COM displacement during the gait cycle (see Fig. 1) revealed a significant group $\times$ task interaction [$F(3, 129) = 5.31$, $p = .004$, $\eta_p^2 = .110$] and a main effect of time [$F(4, 172) = 4.67$, $p = .004$, $\eta_p^2 = .098$]. Pairwise follow-up comparisons indicated that in the MAS and Q&A conditions, the concussion group displayed significantly more total M/L COM displacement than control subjects at each time point throughout the 2 months of testing, but not in the WALK or SAS conditions. Additionally, for the concussion group, there was significantly less total medial/lateral COM displacement in the WALK condition than for any of the other three conditions and significantly more total medial/lateral COM displacement in the Q&A condition compared with the WALK, SAS, and MAS conditions. No significant differences were found between the four conditions for the control group. Pairwise follow-up comparisons for the main effect of time revealed significantly greater total medial/lateral COM displacement at the

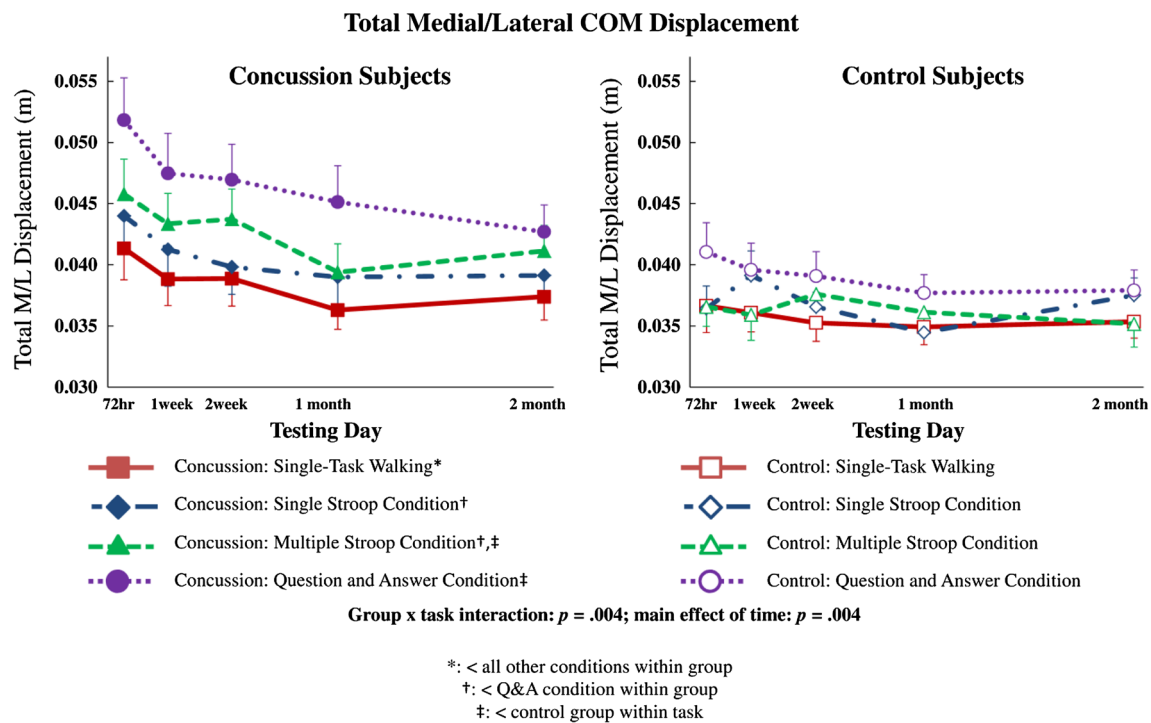


Fig. 1 Mean $\pm$ SE results across the 2-month testing period for total M/L displacement for both groups in each of the four walking conditions. Results indicated a significant group $\times$ task interaction and main effect of time

72-h assessment compared to the 1-month assessment in both subject groups.

Peak M/L COM velocity

Peak medial/lateral COM velocity analysis revealed no significant three-way or two-way interactions ($p > .125$), but detected a main effect of task [see Fig. 2: $F(3, 129) = 11.19$, $p < .001$, $\eta_p^2 = .206$]. Follow-up pairwise comparisons indicated that both groups of subjects demonstrated significantly less peak M/L COM velocity in the WALK, SAS, and MAS conditions compared with the Q&A condition ($p < .01$). Main effects of group [$F(1, 43) = 3.06$, $p = .088$, $\eta_p^2 = .066$] and time [$F(4, 172) = 2.41$, $p = .052$, $\eta_p^2 = .053$] were not statistically significant.

Peak anterior COM velocity

Three separate two-way interactions were detected for the peak anterior COM velocity (see Fig. 3). For the group $\times$ time interaction [$F(4, 172) = 4.44$, $p = .008$, $\eta_p^2 = .094$], results indicated that the concussion group walked with a smaller peak anterior COM velocity within 72 h of injury compared with all subsequent testing times and a smaller peak anterior COM velocity at the 1-week and 2-week assessment in comparison to the 2-month assessment. Control subjects walked with less peak COM

anterior velocity at the initial testing assessment compared with the 1-week testing assessment. The group $\times$ task interaction [$F(3, 129) = 3.80$, $p = .027$, $\eta_p^2 = .081$] analysis indicated significant peak COM anterior velocity differences for concussion subjects between all four conditions (WALK > SAS > MAS > Q&A), while control subjects demonstrated significantly greater peak COM anterior velocity in the WALK compared with the SAS, MAS, or Q&A conditions and significantly greater peak COM anterior velocity in the SAS than the MAS or Q&A conditions. The time $\times$ task interaction [$F(12, 516) = 5.27$, $p < .001$, $\eta_p^2 = .109$] results indicated that in the WALK condition, all subjects displayed a smaller peak COM anterior velocity at the initial testing time compared with the 2-week, 1-month, and 2-month follow-up and a larger peak COM anterior velocity at the 2-month visit compared with all other testing times. For the SAS, MAS, and Q&A conditions, all subjects walked with a significantly smaller peak COM anterior velocity at the first visit compared with all other time points. For the Q&A condition, the 1-week and 2-week follow-up visits were significantly less than the 1- and 2-month visits.

Average walking speed

Three separate two-way interactions were detected for average walking speed (see Fig. 4). The group $\times$ time interaction

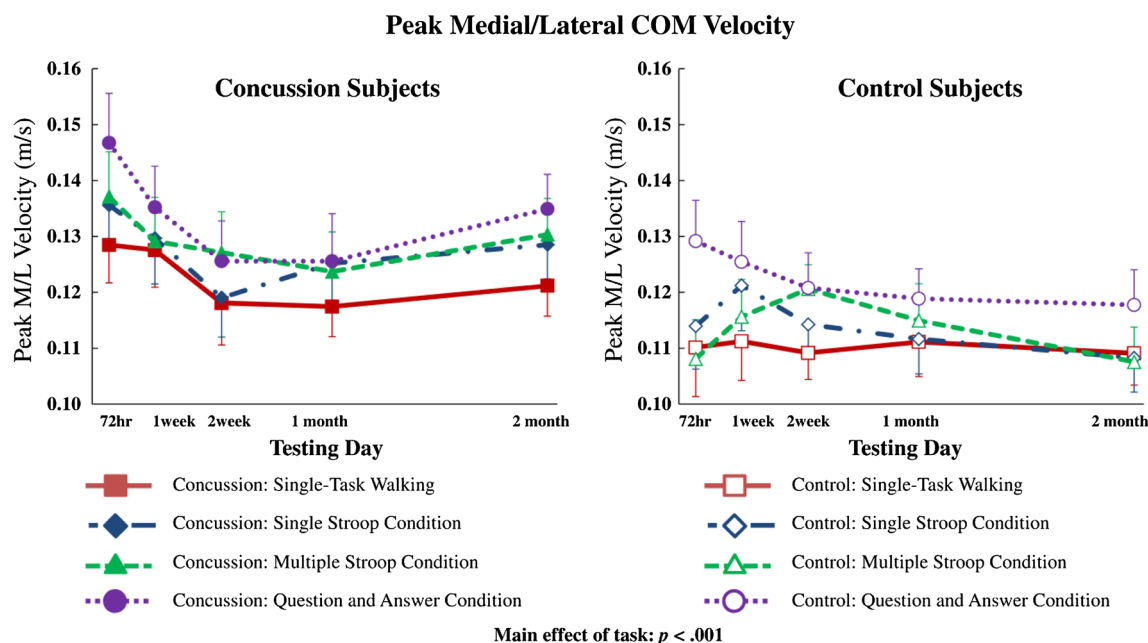


Fig. 2 Mean $\pm$ SE results across the 2-month testing period for peak M/L velocity for both groups in each of the four walking conditions. Results indicated a significant main effect of task with the Q&A

condition displaying significantly greater peak M/L velocity than all other conditions

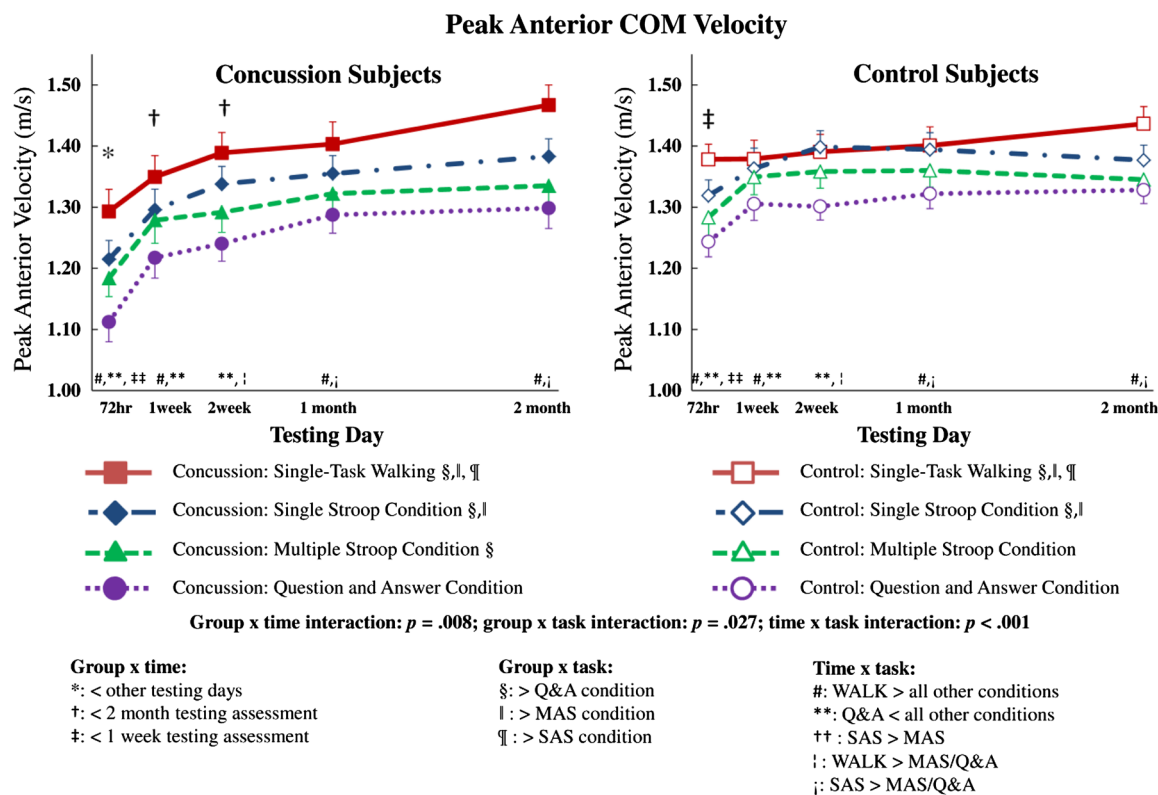


Fig. 3 Mean $\pm$ SE results across the 2-month testing period for peak anterior velocity for both groups in each of the four walking conditions. Results indicated significant group $\times$ time, group $\times$ task, and time $\times$ task interactions

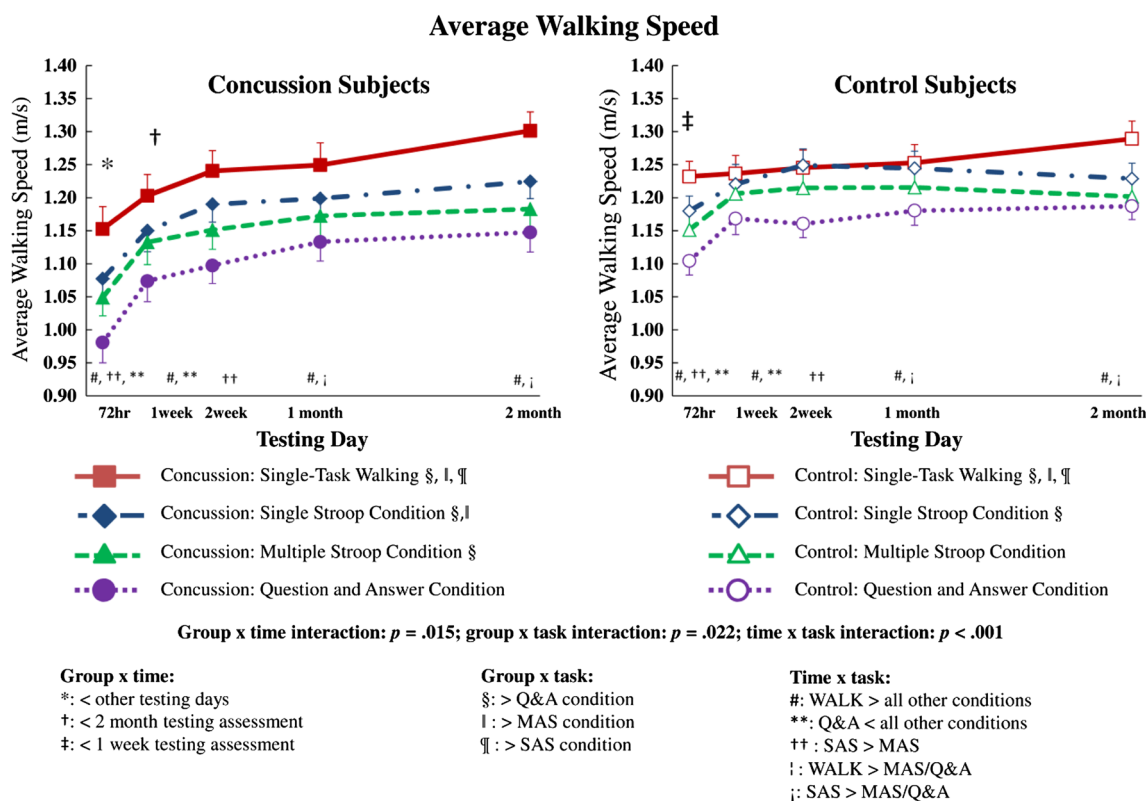


Fig. 4 Mean $\pm$ SE results across the 2-month testing period for average walking speed for both groups in each of the four walking conditions. Results indicated significant group $\times$ time, group $\times$ task, and time $\times$ task interactions

[$F(4, 172) = 3.98, p = .015, \eta_p^2 = .085$] indicated that for the concussion group, the 72 h post-injury visit was significantly slower than all other testing days, the 1-week visit was significantly slower than the 2-month visit, and for the control group, the initial visit was significantly slower than the 1-week follow-up. For the group $\times$ task interaction [$F(3, 129) = 4.11, p = .022, \eta_p^2 = .087$], no differences between groups on any task were detected, but the concussion group demonstrated significant differences between each condition (WALK > SAS > MAS > Q&A), while the control group walked with a faster average walking speed in the WALK condition compared with the other three conditions and a greater walking speed in the SAS condition compared with the Q&A condition. The time $\times$ task interaction [$F(12, 516) = 5.52, p < .001, \eta_p^2 = .114$] revealed the average walking speed for subjects in the WALK condition was less at the 72-h visit in comparison with the 1-month visit, and the 2-month visit was significantly faster than all other visits. For the SAS, MAS, and Q&A conditions, the 72-h visit was significantly slower than all other visits.

Secondary task accuracy

Analysis of the secondary cognitive task accuracy revealed a three-way (group $\times$ time $\times$ task) interaction [see Table 2;

$F(8, 344) = 2.728, p = .015, \eta_p^2 = .006$]. Follow-up comparisons indicated that concussion subjects were significantly less accurate than control subjects on the MAS task at the 72-h, 1-month, and 2-month testing time points and significantly less accurate on the SAS task at the 1-month testing time point. Concussion subjects were also more accurate on the SAS task at the 2-month testing assessment compared with the 72-h and 1-month time points and less accurate at the 72-h time point than the 2-week time point. Further, during the Q&A task, concussion subjects were more accurate at the 2-month assessment than the 72-h, 1-week, and 2-week testing time points. Finally, both groups of subjects were less accurate on the Q&A task than the SAS or MAS tasks at the 72-h, 1-week, 2-week, or 1-month time points. At the 2-month time point, concussion subjects were more accurate on the SAS task than the MAS task and control subjects were more accurate on the SAS and MAS tasks than the Q&A task.

Discussion

This longitudinal and prospective investigation on dynamic balance control indicated that gait stability is affected by task complexity for adolescents following concussion.

Table 2 Cognitive test accuracy rates for both groups of subjects

Task	Group	72 h	1 week	2 weeks	1 month	2 months
Single auditory Stroop (SAS)	Concussion	92.6 % (± 10.5) ^{†‡}	97.7 % (± 5.6)	98.7 % (± 3.5)	97.3 % (± 5.7) ^{*†}	99.5 % (± 2.6)
	Control	96.9 % (± 8.5)	97.1 % (± 8.7)	99.6 % (± 2.1)	100.0 % (± 0.0)	100.0 % (± 0.0)
Multiple auditory Stroop (MAS)	Concussion	96.2 % (± 4.7) [*]	96.9 % (± 4.3)	97.9 % (± 4.5)	98.0 % (± 3.8) [*]	97.6 % (± 3.0) ^{*¶}
	Control	99.2 % (± 1.7)	98.7 % (± 2.3)	99.7 % (± 1.0)	99.7 % (± 1.0)	99.3 % (± 1.5)
Question and answer (Q&A)	Concussion	80.8 % (± 13.2) ^{†§}	82.6 % (± 14.6) ^{†§}	83.1 % (± 15.9) ^{†§}	83.8 % (± 17.8) [§]	91.9 % (± 12.3)
	Control	79.5 % (± 17.4) [§]	88.0 % (± 13.0) [§]	86.8 % (± 13.3) [§]	88.5 % (± 10.5) [§]	84.4 % (± 16.7) [§]

* Concussion group significantly lower than control group

† Significantly <2-month assessment

‡ Significantly <2-week assessment

§ Significantly less than SAS or MAS condition

¶ Significantly less than SAS condition

These findings are in agreement with previous work indicating that the addition of a secondary task while walking appears to affect motor ability in young adults who suffered a concussion (Parker et al. 2006) and adults who have a history of concussion (Fait et al. 2013). The results are similar to a previous report which identified that the complexity levels of a task have a significant effect on gait in children suffering from post-severe TBI (Katz-Leurer et al. 2011). In the present study, the total M/L displacement differences observed between concussion and control groups across 2 months of testing in the two most complex conditions (MAS and Q&A) and within the different conditions for the concussion group suggest that in subjects with concussion, as cognitive task complexity increases, dynamic balance control deficits increase.

The findings suggest that dual-task activities, which require complex integration, prioritization, and execution of motor and cognitive demands (Yogev-Seligmann et al. 2008), may amplify balance disturbances in individuals suffering from concussion. Such disturbances may be reflective of incomplete physiologic recovery even though clinical symptoms may be normal (Barr et al. 2012). Further, others have used dual-task protocols and reported a regression in gait stability from a 2-week assessment to a 1-month assessment following concussion in a group of young adults who returned to pre-injury levels of activity within 2 weeks of injury. It was suggested that motor ability may not be sufficiently recovered despite the clinical observation of recovery (Parker et al. 2006). Subjects in the current study returned to physical activity at an average of 27 days after injury, and the mean medial/lateral velocity value increased in all walking conditions from the 1-month to the 2-month time period for the concussion group but not the control group. This may be indicative of incomplete recovery of dynamic balance control, which may not be detectable through current concussion assessment paradigms.

Neural recruitment has been reported to be the allocation of available network components based upon task demands (Bryer et al. 2013), and the maintenance of balance has been suggested to require a significant amount of information processing capacity (Lajoie et al. 1993). Both neural recruitment and information processing abilities are required to complete a dual task, and it has been documented that concussion may affect the ability to allocate sufficient attention to each component of a dual task, thereby potentially resulting in the disruption of balance control (Catena et al. 2006). The data in the current study suggest these disruptions may be particularly evident under complex dual-task conditions.

The results demonstrate that individuals suffering from concussion display greater deficits to gait stability in more complex conditions, while cognitive task performance differences between groups were not seen consistently across the testing time period. Hence, performance during complex dual tasks may result in greater gait instability compared to dual tasks of lesser complexity, potentially affected by the difficulty in information processing ability (Bernstein 2002) or processing speed (Fazio et al. 2007) following concussion, resulting in increased motor difficulty during a complex dual task.

Previous reports have indicated that instability while walking can be detected using the motion and velocity of the COM in the frontal plane (Hahn and Chou 2003). In the current study, however, the peak COM M/L velocity did not distinguish between groups or across time, but did between tasks suggesting that in individuals with concussion particularly, this variable may be sensitive to task complexity.

Similar to the effect of task complexity on total COM M/L displacement, peak COM anterior velocity also appeared to be sensitive to different levels of dual-task complexity in the concussion group. Peak COM anterior velocity has previously been documented as an indicator of

forward momentum control in accommodating to divided attention (Catena et al. 2006; Howell et al. 2013b). In this study, adolescents with concussion displayed a decrease in peak COM anterior velocity as cognitive task complexity increased, while control subjects did not demonstrate differences in peak COM anterior velocity in the most complex conditions (MAS and Q&A). Thus, forward momentum control during dynamic tasks, reported as a continuous regulation and integration of sensory inputs (Lajoie et al. 1993), may be highly affected by task complexity for adolescents with concussion.

Disruptions to cognition, such as attention and task switching ability, have been documented for a time period of up to 2 months following a concussion for an adolescent population (Howell et al. 2013a). During this time of life, environmentally induced changes in the cortex may be detrimental to the continued development of the brain (Crews et al. 2007), making the adolescent population particularly susceptible to the effects of head trauma. Concussion-induced motor cortex disruption has been well documented following injury (Henry et al. 2011; Tallus et al. 2012; De Beaumont et al. 2012) and may be a contributor to gait imbalance following concussion.

Proper attentional distribution is considered especially needed to perform multiple tasks when the system is near its limits (Yogev-Seligmann et al. 2008) and the failure to flexibly switch between tasks or to distribute attention across tasks may occur in those suffering from brain trauma (Dreher and Grafman 2003). As attentional disruptions may be particularly susceptible to the effects of concussion (van Donkelaar et al. 2005) and individuals suffering from concussion have difficulties in orienting attention (Drew et al. 2007) or other decreased attentional abilities (Ma et al. 2012), increased task complexity may require prioritization; this may result in the inability to complete both tasks with equal success.

Dual tasks have been indicated in the management of concussion (Teel et al. 2013) and reported as a way to detect subtle neurologic deficits associated with concussion (Lee et al. 2013). If motor and cognitive capacities are limited, at least one of them may deteriorate when performed simultaneously (Yogev et al. 2005). As the applicability of dual-task assessments in the clinical management of concussion has recently been documented as a way to increase sensitivity to subtle deficits following injury (Ross et al. 2011), this type of testing may provide an additional tool through which accurate assessment of recovery from concussion can occur. Further, the utilization of more complex tasks in dual-task scenarios may be able to better detect subtle neurologic deficits beyond traditional testing techniques, and more sensitively identify complete recovery following concussion.

Limitations

While the prospective and longitudinal design of this study allows for the monitoring of an acutely injured individual from a time period immediately after the injury and through the 2 months following the injury, the severity of injury for each individual may have varied highly. Previous reports have included concussion grading as an attempt to quantify injury severity (Cantu 2001); however, more recent literature has abandoned grading scales in order to individually guide return to activity decisions following injury (McCrory et al. 2005). The study was also limited as no subject baseline data were available for analysis. Consistent with previous work (Howell et al. 2013a), the inclusion of individually matched healthy control subjects to which the sample of concussion subjects was compared to for the same testing protocol and timeline countered this limitation to some extent.

Conclusion

Dual-task complexity appears to affect gait balance control to a greater degree in adolescents recovering from concussion than a similarly matched group of control subjects. Varying dual-task complexity represents a useful way to identify motor or cognitive recovery following concussion and may assist clinicians in the management and recovery monitoring following concussion.

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